
Software Requirements Specification

for

Result Archive System

Version 1.1

Prepared by

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Course:	CSE 4211 Software Engineering
Date:	07/01/2025

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Arup Roy, Ashrafur Rahman Ador, Saif Bashar	Initial Draft	03/01/2025
1.1	Arup Roy, Ashrafur Rahman Ador, Saif Bashar	Include Use Case Diagram	05/01/2025

1 Introduction

1.1 Document Purpose

The **Result Archive System (RAS)** is a web-based program intended to enhance the management and accessibility of academic results. The system offers a secure, role-specific platform for three categories of users: Students, Coordinators, and Super Admins.

- **Students** may see their scores and personal information by logging in.
- **Coordinators** are tasked with overseeing academic outcomes, which encompasses the addition, modification, and examination of all students' results.
- **Super Admins** supervise the entire system, administering users (students and coordinators) and executing all actions permitted to other positions.

The objective of this system is to guarantee:

- Prompt and secure access to academic outcomes.
- Effective administration of outcome data.
- Implementation of role-based access control to ensure data integrity and security.
- Minimization of manual inaccuracies in outcome management.

This SRS paper delineates the requirements, functions, and technical specifications essential for constructing an efficient Result Archive System.

1.2 Product Scope

The Result Archive System offers a single platform for the administration of academic results inside an educational institution. The system's objectives are:

- Substitute manual result management with an automated digital procedure.
- Establish a safe platform for students to retrieve their findings.
- Enable coordinators to effectively manage academic data.
- Authorize super administrators to monitor user access and administer the complete system.

The system's principal features comprise:

- Role-Based Authentication:
 - a. **Students** are permitted to access their individual results and information.
 - b. **Coordinators** are able to examine, add, and amend outcomes for all students.
 - c. **Super Admins** possess the authority to oversee users and outcomes, encompassing the ability to add or remove people.
- Data Integrity and Privacy:
 - a. Access to results is restricted to authorized users only.

- b. Safeguarded retention of user credentials and outcome data.
- Scalability:
 - a. The system is engineered to accommodate an increasing user base and output as the institution grows.
- Accessibility:
 - a. Available on multiple devices, including PCs, laptops, tablets, and smartphones.

The system is essential for maintaining transparency and accountability in the management of academic records.

1.3 Intended Audience and Document Overview

This document is intended for multiple stakeholders:

1. **Developers:** To understand the requirements and design implementation strategies.
2. **Testers:** To develop test cases and validation scenarios.
3. **Project Managers:** To coordinate development timelines and deliverables.
4. **Stakeholders:** To review and approve the functional and non-functional requirements.
5. **End Users:** As a reference for understanding system features and benefits.

1.4 Definitions, Acronyms and Abbreviations

- **RAS:** Result Archive System.
- **SRS:** Software Requirements Specification.
- **Student:** A user who can only view their own results and personal details.
- **Coordinator:** A user responsible for managing academic results for students.
- **Super Admin:** A user with the highest privileges, responsible for managing all users and system functionalities.
- **Authentication:** The process of verifying user credentials to access the system.
- **Role-Based Access Control (RBAC):** An approach to restricting system access based on user roles.

1.5 Document Conventions

- **Bold** text is used for headings and key points.
- **Italic** text is used to emphasize specific terms or conditions.
- All dates follow the **YYYY-MM-DD** format.
- **Tables and bullet** points are used to organize information clearly.

1.6 References and Acknowledgments

- IEEE Software Requirements Specification (SRS) Template.
- Institutional Academic Policy for result management.
- Data Protection Regulations (e.g., GDPR, FERPA) applicable to educational institutions.

- Technological Frameworks: Guidance on web-based application development.

2 Overall Description

2.1 Product Overview

The Result Archive System functions as an independent web application for the maintenance of academic results. The technology is designed to supplant conventional result management with a digital, automated procedure. It assimilates into the institution's current IT framework, guaranteeing data security and user-friendliness.

The system accommodates three user roles:

1. **Students:** Restricted access to individual academic information.
2. **Coordinators:** Oversight of academic performance for all pupils.
3. **Super Admins:** Complete authority over people and data.

The solution guarantees

- **Centralized data management:** By securely storing all results and user data in a central database.
- **Role-Based Access Control:** Mitigates unlawful access and safeguards data privacy.
- **Audit Trails:** Records all user activity for accountability and diagnostic purposes.

2.2 Product Functionality

Student Functions:

- Log in to the system using credentials.
- View personal results, including grades, marks, and subject details.
- View personal details, such as name, ID, and academic program.
- Students cannot modify or delete any data.

Coordinator Functions:

- Log in to the system using credentials.
- View results of all students in the system.
- Add new results for students, including subject details, grades, and marks.
- Modify existing results in case of errors or changes.
- Coordinators do not have access to user management functionalities.

Super Admin Functions:

- Log in with the highest-level credentials.
- Add, edit, and delete user accounts (Students and Coordinators).
- View, add, and modify results for all students.
- Perform all actions available to Students and Coordinators.
- Maintain system settings and oversee data security.

2.3 Design and Implementation Constraints

- **Data Format Compatibility:** Must support PDFs, pictures, and spreadsheets.
- **Access Control:** Established responsibilities for administrators, faculty, and students.
- **Search Functionality:** Effective search and filtering utilizing various criteria.
- **Performance:** Must accommodate a minimum of 500 concurrent users, with scalability to 2000 users.
- **Security Protocols:** Data encryption and secure authentication are mandatory.
- **Integration:** Required to interface with current Learning Management Systems.
- **Usability:** Intuitive UI, compatible with both mobile and desktop platforms.
- **Compliance:** Adhere to legal data privacy rules, such as FERPA.
- **Maintenance:** Systematic updates and backups should be scheduled.
- **Financial Limitations:** Aggregate expenses must remain within designated budgetary confines.
- **Timeline:** Development must synchronize with the academic calendar deadlines.

2.4 Operating Environment

Platform: A web-based application accessible using contemporary browsers including Chrome, Firefox, and Edge.

Devices: Compatible with desktop computers, laptops, tablets, and smartphones.

Hosting: Cloud-based infrastructure (e.g., AWS, Azure).

Database: Relational database for the storage of user and result data (e.g., MySQL, PostgreSQL).

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

- **Login Page:** Allows secure authentication.
- **Student Dashboard:** Displays personal results and details.
- **Coordinator Dashboard:** Displays tools for managing results.
- **Super Admin Dashboard:** Provides user and result management functionalities.

3.1.2 Hardware Interfaces

The system should run on devices with standard hardware configurations (e.g., 4GB RAM, modern processors).

3.1.3 Software Interfaces

- **Backend:** Node.js or Python (Django/Flask).
- **Frontend:** React.js or Angular.
- **Database:** MySQL, PostgreSQL.

3.2 Functional Requirements

3.2.1 User Authentication

- Users must log in with a valid username and password.
- Role-based access control limits features based on user roles.

3.2.2 Student Functionalities

Students can view:

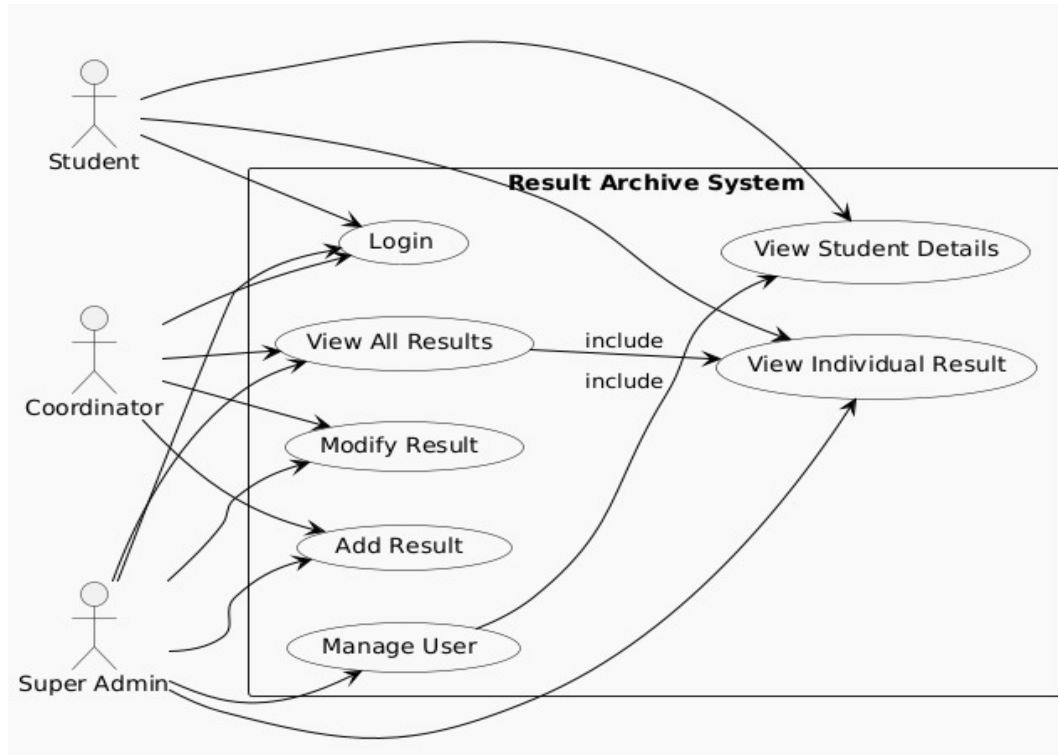
- Their personal academic results.
- Their personal details (e.g., name, ID, email).
- No permissions to modify or delete data.

3.2.3 Coordinator Functionalities

Coordinators can:

- View results for all students.
- Add new results, including subject, marks, grades, and academic year.
- Modify incorrect or outdated results.

3.3 Use Case Model



4 Other Non-functional Requirements

4.1 Performance Requirements

- Support up to 500 concurrent users.
- Actions should execute within 3 seconds.

4.2 Safety and Security Requirements

- Passwords must be encrypted and stored securely.
- Data must be transmitted using HTTPS.
- Role-based access control ensures data privacy.
- Audit logs track all user actions.

4.3 Software Quality Attributes

- **Scalability:** Should support future feature additions.
- **Usability:** User-friendly interfaces for all roles.

5 Other Requirements

System Performance: Response time for queries should not exceed 2 seconds.

Data Encryption: All sensitive data must be encrypted at rest and in transit.

User Training: Comprehensive training sessions must be provided for all users upon system launch.

Reporting Capability: Must generate customizable reports on student performance and archiving statistics.

User Support: Provide a helpdesk for user inquiries and technical support.

Mobile Access: The system must be accessible via mobile devices with a responsive design.

Browser Compatibility: Support for major browsers (Chrome, Firefox, Safari, Edge).

Localization: The system must support multiple languages based on user preferences.

Version Control: Maintain version history for all archived results to track changes.

System Integration: Compatibility with third-party educational tools and APIs for data exchange.

Appendix A – Demo Account

Super Admin: super@gmail.com / password

coordinator: admin@gmail.com / password

Students: user@gmail.com / password

Appendix B - Group Log

Date	Action Taken	User Id	Description
03/01/2025	SRS	04,10,11	Gather information from the client and make a initial draft of SRS
05/01/2025	Use Case Diagram	04,10,11	After drafting a SRS document we make a use case diagram